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grep

Definition

grep (Global Regular Expression Print) is used to search for specific patterns or text inside files or command output.

Usage / Formula

```
grep [options] "pattern" file
```

Examples

```
grep "error" system.log           # Find lines containing "error"
grep -i "error" system.log        # Case-insensitive search
grep -n "user" access.log         # Show line numbers
```

awk

Definition

awk is a powerful text-processing tool used for pattern scanning, column extraction, and reporting.

Usage / Formula

```
awk 'pattern {action}' file
```

Examples

```
awk '{print $1}' file.txt          # Print first column
awk '{print $3, $4}' file.txt      # Print columns 3 and 4
awk '$3 == "user42"' file.txt     # Filter rows where column 3 matches
```

sed

Definition

sed (Stream Editor) is used to modify text in files or streams (search, replace, delete).

Usage / Formula

```
sed 's/old/new/' file
```

Examples

```
sed 's/error/success/' file.txt      # Replace first occurrence per line
sed 's/error/success/g' file.txt     # Replace all occurrences
sed -n '5,10p' file.txt              # Print lines 5 to 10
```

Pipe Operator |

Definition

The pipe (|) sends the output of one command as input to another command.

Usage / Formula

```
command1 | command2
```

Examples (3 required)

```
cat file.txt | grep "error"          # Search inside file output
ls -l | grep ".txt"                  # Filter .txt files
cat access.log | awk '{print $3}'    # Extract usernames
```

Output Redirection >

Definition

> saves the output of a command into a file (overwrites existing content).

Usage / Formula

```
command > file.txt
```

Examples (3 required)

```
ls > files.txt           # Save directory listing
cat file.txt > copy.txt   # Copy file content
echo "Hello" > greeting.txt # Write text to file
```

Append Redirection >>

Definition

>> appends output to a file without deleting existing content.

Usage / Formula

```
command >> file.txt
```

Examples (3 required)

```
echo "New line" >> file.txt # Add line to file
date >> log.txt             # Append date to log file
ls >> files.txt             # Append directory listing
```
